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Submitted by email to ostp-ai-rfi@nitrd.gov

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan ("Plan")

Date: March 10, 2025

Subject: AI Action Plan

Statement:
This document is approved for public dissemination and contains no business-proprietary or confidential information. The government may reuse its contents in developing the AI Action Plan and associated documents without attribution.

Summary:
LavaCrypt, a research and development company specializing in cybersecurity, AI, and quantum computing technologies, commends the Office of Science and Technology Policy (OSTP) and the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO) for soliciting public input on the crucial development of an Artificial Intelligence (AI) Action Plan. A robust and forward-thinking AI Action Plan is essential for maintaining U.S. leadership in AI, fostering economic growth, ensuring national security, and promoting human flourishing. Our recommendations emphasize a balanced approach that prioritizes innovation and responsible development, focusing on secure AI development, workforce readiness, strategic international

collaboration, and addressing potential risks while avoiding overly burdensome regulations. Our suggestions will keep the USA at the top of AI development.

1. Introduction

LavaCrypt appreciates the opportunity to contribute to developing the AI Action Plan. As a company deeply involved in cutting-edge AI research and development, particularly in its intersection with cybersecurity and quantum computing, we understand AI's transformative potential and the critical need for a comprehensive national strategy. This plan should focus on advancing AI technology and anticipating and mitigating potential domestic and international risks.

2. Priority Policy Actions for the AI Action Plan

LavaCrypt recommends the following priority policy actions for inclusion in the AI Action Plan:

2.1. Secure AI Development & Deployment

2.1.1. AI Cybersecurity Standards

The establishment of mandatory cybersecurity standards for AI systems requires a comprehensive framework that addresses multiple security layers:

Data Provenance and Integrity

- Implement cryptographic signatures and blockchain verification to establish immutable audit trails for training datasets.
- Develop standardized metadata requirements, documenting data sources, collection methodologies, and processing history.
- Establish centralized repositories for verified, high-quality training data with strict access controls.
- Create certification programs for organizations that maintain exemplary data governance practices.

Model Robustness and Resilience

- Mandate stress testing against common adversarial attacks (e.g., evasion attacks, model inversion).
- Establish minimum performance thresholds under various perturbation scenarios.
- Implementation of defensive techniques such as adversarial training and robust optimization is required.
- Develop quantitative metrics for measuring model resilience against specific attack vectors.

Explainability and Auditability

- Define tiered explainability requirements based on application risk categories.
- Establish standardized documentation protocols for model architectures and training methodologies.
- Implement interpretability tools appropriate to the model type (e.g., SHAP values, integrated gradients).
- Create certification programs for third-party auditing organizations.

Secure AI Lifecycle Management

- Establish version control requirements for models, datasets, and related artifacts.
- Implement automated security scanning throughout the development pipeline.
- Define secure procedures for model updates and parameter tuning.
- Create mandatory incident response protocols specific to AI systems.

Vulnerability Disclosure and Patching

- Establish a standardized vulnerability scoring system specific to AI systems.
- Create centralized vulnerability reporting platforms with appropriate protections for researchers.
- Define maximum timeframes for addressing critical vulnerabilities.
- Implement security update verification protocols to prevent exploitation of the patching process.

2.1.2. AI Red Teaming and Penetration Testing

An effective red team program for AI systems should include:

- Formation of specialized AI security assessment teams with cross-disciplinary expertise.
- Development of standardized testing methodologies for different AI system types.
- Creation of a national certification program for AI security testers.
- Establishment of a secure information-sharing framework for discovered vulnerabilities.
- Implementation of progressive financial incentives based on vulnerability severity.
- Development of specialized testing environments that simulate real-world deployment conditions.
- Creation of a national leaderboard recognizing top security researchers in the AI space.
- Regular publication of anonymized findings to benefit the broader ecosystem.

2.1.3. Research into AI Safety and Security

The OSTP/NITRD funding program should prioritize:

Adversarial Machine Learning

- Research into novel attack vectors specific to large language models and multimodal systems.
- Development of mathematical frameworks for quantifying model vulnerability.
- Creation of standardized benchmarks for evaluating defense mechanisms.
- Investigation of transferability of adversarial examples across different model architectures.

Formal Verification of AI Systems

- Development of verification methodologies for neural network architectures.
- Research tractable approaches for verifying properties of large-scale models.

- Creation of automated tools for continuous verification during model development.
- Standardization of formal property specifications for different AI application domains.

Differential Privacy Applied to AI Models

- Research on optimal privacy budget allocation across model training processes.
- Development of techniques that balance privacy guarantees with model utility.
- Creation of industry-specific guidelines for privacy parameter selection.
- Investigation of composition effects in complex AI pipelines.

Encryption and Privacy-Enhancing Technologies

- Research on computational efficiency improvements for full encryption.
- Development of specialized hardware accelerators for privacy-preserving computation.
- Creation of privacy-preserving federated learning frameworks.
- Investigation of secure multi-party computation techniques for model training.

2.2. Workforce Development and Education

2.2.1. AI Skills Gap Initiative

The national initiative should implement the following:

Curriculum Development

- Create age-appropriate AI education modules for K-12 that integrate with existing subjects.
- Develop standardized undergraduate curricula covering core AI competencies.
- Establish specialized graduate programs addressing emerging AI subfields.
- Create vocational training programs focused on practical AI implementation skills.

- Develop micro-credential programs targeting specific industry applications.

Teacher Training

- Establish summer institutes for K-12 educators to develop AI teaching competencies.
- Create online certification programs for educators at all levels.
- Develop comprehensive teaching resources and lesson plans.
- Form partnerships with industry to provide real-world context and examples.
- Create teacher-mentor networks connecting experienced AI educators with newcomers.

Scholarships and Fellowships

- Establish tiered scholarship programs targeting different educational levels.
- Create industry-sponsored fellowships with practical work components.
- Develop research fellowships focused on specific national priority areas.
- Implement service-based scholarships with post-graduation public sector commitments.
- Establish re-entry scholarships for professionals changing careers.

Reskilling and Upskilling Programs

- Develop industry-specific training programs targeting high-demand sectors.
- Create accelerated certification programs for professionals with adjacent skills.
- Establish tax incentives for employers investing in employee AI skill development.
- Implement regional training centers in areas impacted by technological displacement.
- Develop online learning platforms with personalized skill development pathways.

Promote Diversity and Inclusion

- Establish targeted outreach programs for underrepresented communities.

- Create mentorship networks connecting diverse AI professionals with students.
- Implement early intervention programs in underserved educational districts.
- Develop scholarship programs specifically targeting diversity gaps.
- Create industry partnerships focused on diversifying AI talent pipelines.

2.2.2. AI Ethics Training

A comprehensive ethics component should include:

- Development of case-based learning modules drawn from real-world ethical dilemmas.
- Creation of simulation environments for ethical decision-making practice.
- Establishment of cross-disciplinary ethics coursework drawing from philosophy, sociology, and law.
- Implementation of certification programs in responsible AI development.
- Development of industry-specific ethical guidelines and best practices.
- Creation of ethics coaching and consulting resources for organizations.
- Establishment of professional standards of conduct for AI practitioners.

2.3. International Collaboration and Competition

2.3.1. Strategic Partnerships

Effective strategic partnerships require:

- Formation of multinational research consortia focused on shared priority areas.
- Establishment of talent exchange programs between allied nations.
- Development of shared funding mechanisms for collaborative research initiatives.
- Creation of standardized intellectual property frameworks for joint innovations.
- Implementation of secure information-sharing protocols for sensitive research.

- Formation of industry-led international working groups on standards and best practices.
- Development of shared testing and evaluation facilities.
- Establishment of joint academic programs between leading institutions.

2.3.2. AI Diplomacy

A robust AI diplomacy approach should include the following:

- Creation of dedicated AI attaché positions within key embassies.
- Establishment of multilateral forums explicitly focused on AI governance.
- Development of international principles for responsible AI development.
- Formation of rapid response protocols for addressing emerging AI risks.
- Implementation of capacity-building programs for developing nations.
- Creation of shared regulatory frameworks with close allies.
- Development of international incident response mechanisms.
- Establishment of global AI ethics observatories.

2.3.3. Export Controls

A practical export control framework should feature the following:

- Development of tiered control categories based on potential dual-use applications.
- Creation of streamlined licensing processes for trusted partners and allies.
- Establishment of regular review cycles to adapt to technological evolution.
- Implementation of post-export monitoring mechanisms for critical technologies.
- Development of industry compliance assistance programs.
- Formation of multilateral control regimes with like-minded nations.
- Creation of specialized technical advisory committees for control list updates.
- Establishment of expedited processes for humanitarian and research applications.

2.4. Fostering Innovation and Avoiding Burdensome Regulations

2.4.1. Regulatory Sandboxes

Effective sandbox implementation includes:

- Creation of sector-specific sandboxes with tailored regulatory modifications.
- Establishment of clear entry criteria and operational boundaries.
- Development of robust monitoring and evaluation frameworks.
- Implementation of graduated risk management protocols.
- Formation of multi-stakeholder governance committees.
- Creation of knowledge-sharing mechanisms between participants.
- Establishment of clear pathways to full regulatory compliance.
- Development of expedited approval processes for successful sandbox graduates.

2.4.2. Principles-Based Regulation

A principles-based regulatory approach should:

- Establish clear, outcome-focused regulatory objectives.
- Define flexible implementation pathways suitable for diverse organizations.
- Develop tiered compliance requirements based on application risk profiles.
- Create robust accountability mechanisms without prescribing specific technologies.
- Implement regular stakeholder consultation processes.
- Establish safe harbor provisions for good-faith compliance efforts.
- Develop industry-specific guidance documents and best practices.
- Create certification programs for regulatory compliance.

2.4.3. Public-Private Partnerships

Effective partnerships require:

- Development of clear intellectual property frameworks for joint innovations.
- Establishment of shared research facilities with flexible access models.
- Create coordinated funding mechanisms that blend public and private resources.
- Implementation of streamlined contracting vehicles for rapid collaboration.
- Formation of industry consortia focused on pre-competitive research.
- Development of data-sharing agreements with appropriate privacy safeguards.
- Establishment of researcher exchange programs between sectors.
- Creation of innovation challenges addressing specific national priorities.

2.4.4. Streamlined Approvals

An efficient approval process includes the following:

- Development of risk-based assessment frameworks for prioritizing applications.
- Creation of pre-certification programs for established developers.
- Establishment of expedited review tracks for innovations addressing critical needs.
- Implementation of phased approval processes for iterative deployment.
- Formation of specialized review teams with domain-specific expertise.
- Development of standardized evidence requirements by application category.
- Creation of post-approval monitoring protocols proportionate to risk.
- Establishment of regular process improvement reviews.

2.5. Quantum-Resistant AI

2.5.1. Quantum Computing Threat Assessment

A comprehensive assessment should include:

- Analysis of vulnerability timelines based on quantum computing development projections.

- Identification of critical AI systems potentially susceptible to quantum attacks.
- Development of risk prioritization frameworks for remediation efforts.
- Creation of classified and unclassified assessment versions for different stakeholders.
- Implementation of continuous monitoring of quantum computing advances.
- Formation of specialized assessment teams combining quantum and AI expertise.
- Establishment of secure reporting channels for newly identified vulnerabilities.
- Development of sector-specific vulnerability profiles and mitigation strategies.

2.5.2. Develop and Fund Quantum-Resistant Cryptography

A robust funding program should support:

- Research into lattice-based, hash-based, code-based, and multivariate cryptographic systems.
- Development of standardized implementation libraries for common platforms.
- Creation of transition frameworks for migrating existing systems.
- Establishment of validation programs for proposed solutions.
- Implementation of bug bounty programs specific to quantum-resistant algorithms.
- Formation of specialized research centers focused on post-quantum cryptography.
- Development of hardware acceleration for quantum-resistant approaches.
- Creation of compliance testing frameworks for critical infrastructure.

2.5.3. Quantum-Resistant AI Research

Priority research areas should include:

- Development of learning algorithms with inherent quantum resistance.

- Creation of encryption methods for model parameters and architectures.
- Establishment of secure inference protocols resistant to quantum attacks.
- Implementation of quantum-resistant federated learning approaches.
- Formation of secure model serving architectures with post-quantum protections.
- Development of quantum-resistant privacy-preserving computation techniques.
- Creation of secure update mechanisms for deployed AI systems
- Establishment of certification programs for quantum-resistant AI implementations.

3. Conclusion

LavaCrypt believes that the AI Action Plan represents a critical opportunity to shape the future of AI in the United States. By prioritizing secure AI development, workforce readiness, strategic international collaboration, and responsible innovation, the U.S. can maintain its global leadership in AI and harness the transformative power of this technology for the benefit of all. We urge the OSTP and NITRD NCO to carefully consider these recommendations and incorporate them into the AI Action Plan. We stand ready to assist in any way possible.